

Henry Nitzberg

henry.nitzberg@tufts.edu • (510) 993-6355

Driven and organized with interests in ML, robotics, mathematics, and visual art.

Skills

- **Languages:** Python, C++, C, JavaScript/TypeScript, Java
- **Libraries & Frameworks:** OpenCV, MQTT, GStreamer, GTest, PyTorch, Tensorflow, NumPy, Stable Baselines, IsaacSim, ROS, React, Node/NPM, Mongoose, Ssqlalchemy, FastAPI
- **Miscellaneous:** Docker, CAD, Copilot, Cursor

Experience

Nucleus Biologics | *Digital System and Machine Learning Engineer* San Diego, CA | **July 2025 - Present**

- Built LLM based tools with guardrails for RAG, customer document parsing, and incorporated with inventory information to calculate products' chemical feasibility.
- Managed remote team to optimize internal costing and quoting system.
- Designed and built a custom end-to-end order management system including modules for costing, pricing, resource and supply chain management, compliance, QA, and shipping and receiving.

CoRe Lab | *Research Assistant (Advisor: Yixin Zhu)* Peking University (remote) | **January 2025 - April 2025**

GPU Accelerated Tactile Sensor Simulation

- Wrapped Environment in Gym-compatible class for SB3
- Wrote RL demos using Tacel library meant to integrate & release with the Nvidia Warp simulator

Hayden AI | *Intern, C++ Developer* San Francisco, CA | **June 2024 - September 2024**

Automatic School Bus Stop Arm Violation Detection System

- Built classifier for school bus state (stopped, driving, loading, etc.) from signal data
- Translated GPS data to a human-readable format (UBlox library)
- Created C++ wrapper for ffmpeg bash command to cut and stitch videos at the edge
- Integrated above modules as distinct nodes into a pub sub architecture launchable with Docker

Human Robot Interaction Lab | *Research Assistant, Robotics Engineer* Tufts University | **June 2023–May 2024**

Pathfinding and general assistant robot for hospital patients

- Built robust vision pipeline in python/c++ to detect, identify, and localize indoor elements
- Trained DRL algorithm in custom simulation to optimize for elevator boarding in chaotic environments
- Built ROS package to translate output from DRL model to command velocities
- Developed DIARC component for Fetch, allowing the robot to press elevator call buttons

Art Commissions 2019– 2023

- Graphite, Acrylic, Charcoal, Watercolor

Education

Tufts University GPA: 3.9 | **Fall 2021 - Spring 2025**

B.S. Summa Cum Laude in Computer Science, Mathematics, Minor Studio Art

Multi-LLM Classifier Fall 2025

CS135: Machine Learning and Data Mining course project (with Zeno Marquis)

- Outperformed all state-of-the-art non-LLM sentiment analysis models using ensemble of LLMs (99.8% accuracy)

End to End Self Driving in Simulation (GitHub) Spring 2024

CS141: Probabilistic Robotics for HRI course project (with Clarence Yeh)

- Trained separate DCNNs for image segmentation and control (steering angle, acceleration) using custom data
- Generated, collected and augmented human driving data using CARLA
- Deployed end-to-end self-driving model in various realistic video games

Fan of maker projects, running, climbing, coding, smoothies, Fine Art, Netflix